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Subject: OSSP(PRACTICAL)

1.FCFS Scheduling Algorithms:

```
GNU nano 8.7.1 fcf
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

OUTPUT:

```
vaibhavi@VYSRI: ~/schedulin × + v
vaibhavi@VYSRI:~$ mkdir scheduling
vaibhavi@VYSRI:~$ cd scheduling
vaibhavi@VYSRI:~/scheduling$ pwd
/home/vaibhavi/scheduling
vaibhavi@VYSRI:~/scheduling$ nano fcfs.c
vaibhavi@VYSRI:~/scheduling$ ^C
vaibhavi@VYSRI:~/scheduling$ gcc fcfs.c -o fcfs
vaibhavi@VYSRI:~/scheduling$ ./fcfs
Enter number of processes: 5
Enter burst time for each process:
P1: 5
P2: 4
P3: 3
P4: 1
P5: 2

Process BT      WT      TAT
P1      5      0      5
P2      4      5      9
P3      3      9     12
P4      1     12     13
P5      2     13     15
vaibhavi@VYSRI:~/scheduling$ |
```

2.SJF_Nonpreemptive

```
vaibhavi@VYSRI: ~/schedulin
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort burst times
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

OUTPUT:

```
vaibhavi@VYSRI: ~/schedulin  X + v
vaibhavi@VYSRI:~$ mkdir scheduling
mkdir: scheduling: File exists
vaibhavi@VYSRI:~$ cd scheduling
vaibhavi@VYSRI:~/scheduling$ pwd
/home/vaibhavi/scheduling
vaibhavi@VYSRI:~/scheduling$ nano sjf.c
vaibhavi@VYSRI:~/scheduling$ nano sjf.c
vaibhavi@VYSRI:~/scheduling$ gcc sjf.c -o sjf
vaibhavi@VYSRI:~/scheduling$ ./sjf
Enter number of processes: 5
Enter burst time:
P1: 5
P2: 6
P3: 7
P4: 8
P5: 9

Process  BT      WT      TAT
P1        5        0        5
P2        6        5       11
P3        7       11       18
P4        8       18       26
P5        9       26       35
vaibhavi@VYSRI:~/scheduling$ |
```

3.Priority



vaibhavi@VYSRI: ~/schedulin



GNU nano 8.7.1

```
#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    // Sort according to priority
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (priority[i] > priority[j]) {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t\t%d\t%d\n",
            i + 1, bt[i], priority[i], wt[i], tat[i]);
    }

    return 0;
}
```

```

P5      9      26      35
vaibhavi@VYSRI:~/scheduling$ nano priority.c
vaibhavi@VYSRI:~/scheduling$ gcc priority.c -o priority
vaibhavi@VYSRI:~/scheduling$ ./priority
Enter number of processes: 10
Enter burst time and priority:
P1 Burst Time: 5
P1 Priority: 2
P2 Burst Time: 3
P2 Priority: 1
P3 Burst Time: 4
P3 Priority: 5
P4 Burst Time: 4
P4 Priority: 7
P5 Burst Time: 9
P5 Priority: 8
P6 Burst Time: 6
P6 Priority: 1
P7 Burst Time: 3
P7 Priority: 4
P8 Burst Time: 10
P8 Priority: 9
P9 Burst Time: 10
P9 Priority: 8
P10 Burst Time: 1
P10 Priority: 3

Process BT      Priority      WT      TAT
P1      3      1      0      3
P2      6      1      3      9
P3      5      2      9      14
P4      1      3      14      15
P5      3      4      15      18
P6      4      5      18      22
P7      4      7      22      26
P8      10      8      26      36
P9      9      8      36      45
P10     10      9      45      55
vaibhavi@VYSRI:~/scheduling$ |

```

4.ROUND ROBIN

```
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);

        rem_bt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &quantum);

    // Round Robin
    while (1) {
        done = 1;

        for (i = 0; i < n; i++) {
            if (rem_bt[i] > 0) {
                done = 0;

                if (rem_bt[i] > quantum) {
                    time = time + quantum;
                    rem_bt[i] = rem_bt[i] - quantum;
                }
                else {
                    time = time + rem_bt[i];
```

```
        time = time + rem_bt[i];
        wt[i] = time - bt[i];
        rem_bt[i] = 0;
    }
}

if (done == 1)
    break;
}

// Turnaround time
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n",
        i + 1, bt[i], wt[i], tat[i]);
}

return 0;
}

printf("Enter burst time:\n");

for (i = 0; i < n; i++) {
    printf("P%d: ", i + 1);
    scanf("%d", &bt[i]);

    rem_bt[i] = bt[i];
    wt[i] = 0;
}

printf("Enter time quantum: ");
scanf("%d", &quantum);

// Round Robin
while (1) {
    done = 1;

    for (i = 0; i < n; i++) {
        if (rem_bt[i] > 0) {
            done = 0;

            if (rem_bt[i] > quantum) {
                time = time + quantum;
                rem_bt[i] = rem_bt[i] - quantum;
            }
            else {
                time = time + rem_bt[i];
            }
        }
    }
}
```

```

        }
        else {
            time = time + rem_bt[i];
            wt[i] = time - bt[i];
            rem_bt[i] = 0;
        }
    }

    if (done == 1)
        break;
}

// Turnaround time
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n",
        i + 1, bt[i], wt[i], tat[i]);
}

return 0;
}

```

```

P10      10      9      43      55
vaibhavi@VYSRI:~/scheduling$ nano roundrobin.c
vaibhavi@VYSRI:~/scheduling$ gcc roundrobin.c -o roundrobin
vaibhavi@VYSRI:~/scheduling$ ./roundrobin
Enter number of processes: 4
Enter burst time:
P1: 3
P2: 5
P3: 8
P4: 10
Enter time quantum: 5

Process BT      WT      TAT
P1      3        0        3
P2      5        3        8
P3      8        13       21
P4      10       16       26
vaibhavi@VYSRI:~/scheduling$ |

```